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Group Members: Joseph Wang: jhw45, Aidan Frutos: af314, Isaiah Boyles: isb17, Christian Kirby: cdk37, Jaivir Parmar: jp590

Investigating Energy Usage Patterns in Data-Center Heavy Regions

Part 1: Introduction and Research:

Historically, regional energy consumption has been heavily correlated with weather patterns. Grid demand fluctuated with the seasons, with both summer and winter having higher average energy consumption due to heating and cooling demands. However, with the rapid development of AI and its accompanying data centers, we expect to see a drastic change in energy consumption in data center heavy regions due to the intensive energy requirements for these data centers.

In light of these expected changes, our project aims to investigate the following research questions:

- To what extent has the correlation between weather patterns and grid demand shifted in regions with high data center growth?
- How does the accuracy of energy usage prediction models *that rely heavily on weather features* differ between data center heavy regions and non data center heavy regions?

These research questions are **substantial** because they attempt to answer a question that has not already been definitively answered given how recent and ongoing this change in data center activity as it relates to energy usage is. Both will require work beyond surface-level analysis, as we would be analyzing both the relationship that energy has with weather patterns and with data center usage, and further analyzing what the difference between the two means in each question.

Additionally, they are **feasible** questions due to the fact that we already have a dataset in mind that we can use for each question, and the analysis we are doing can be as deep and as unique as we would like it to be, but even if we hit 60% of the depth of analysis that we could possibly go to, we'd still reach a sufficient answer for this project.

Lastly, they are both definitely **relevant** questions, as the topic of data centers as it relates to energy usage is one of extreme importance right now for various reasons, namely its sustainability, and understanding this relationship in comparison to an existing energy fluctuation that is in our current system would be both interesting and helpful as context to the sustainability issue.

Changes Made:
We have reduced the number of data sources (removed EIA energy data) as it was redundant with the PJM information. We clarified our second research question to focus on models that emphasized weather features (as opposed to models trained on many diverse variables) to see if weather (specifically) is a weaker predictor of energy pricing/demand in data center heavy regions. We elected to use PJM's Day Ahead pricing instead of the 5 minute LMP as we were able to access a longer range of data from this dataset. We also used cleaned yearly load-forecast parquets for system/subregional load, and exported node metadata (zones, coordinates). DA LMP gives a nodal price series we can align to weather; load remains forecast/coarser than a single bus, so node-level "DC-heavy vs not" comparisons lean on LMP.

Part 2: Data Sources:

Our project draws on two primary data sources: weather observations from NOAA's Integrated Surface Database and grid pricing and load data from PJM Interconnection's Dataminer portal. These datasets provide the depth needed to investigate the relationship between weather patterns and energy grid behavior, showing how this has evolved with the rapid expansion of data center infrastructure.

The NOAA NCEI Integrated Surface Database is weather data that was sourced from the National Oceanic and Atmospheric Administration's NCEI Integrated Surface Database. This database provides hourly station-level observations stretching back decades which makes it well-suited for our type of analysis on the weather-energy relationship. Some of the key fields include temperature, dew point, wind speed, atmospheric pressure, and precipitation. In order to align our weather observations with the PJM nodal pricing structure, each PJM node and zone was mapped to the nearest ISD weather station or a weighted blend of the nearby stations. From the station observations, a few engineered features were derived in order to capture the thermal and meteorological signals that influence grid demand, such as Cooling Degree Hours, Heating Degree Hours. These were used to capture HVAC-driven loads.

As for the PJM Interconnection dataminer data, this supplied us grid pricing and load data that were sourced from PJM Interconnection's Dataminer platform. This was selected because it covers the northeastern United States, which has the highest concentration of data centers in the United States. By choosing this region, we figured we would have the most likely data to exhibit divergence between weather-drive and data center-driven energy swings. In total there were three datasets used from the dataminer: *Day-Ahead Locational Marginal Prices, System and Subregional Load Forecasts, and Generation by Fuel Type*. The first is nodal pricing at the day-ahead market level, and we selected day-ahead instead of real-time 5-minute because it offers a longer accessible range. The second was cleaned yearly load-forecast files that were used to capture aggregate and subregional electricity demand. Finally, the third was generation mix data that was incorporated to control for weather-correlated generation sources such as solar and wind output.

Together, these two sources address both of our research questions. The NOAA ISD provides the weather feature set that we need to build and evaluate our models, and the PJM LMP and load data provide the grid outcome variables. This combination of nodal price data and subregional load data allows for analysis at multiple geographic scales, and the deliberate focus on PJM ensures that any detectable shift in weather-energy relationships is most likely to appear in that dataset.

1. NOAA Weather Data - (https://www.weather.gov/documentation/services-web-api)

NOAA (National Oceanic and Atmospheric Administration) is a government agency that focuses on the conditions of the oceans, major waterways, and the atmosphere, including forecasting weather, monitoring climate, and managing marine resources. They provide comprehensive information about weather events within the United States and offer an easily accessible database for the public.

We are focusing entirely on the NOAA NCEI Integrated Surface Database (ISD). This dataset provides a long history, hourly station observations (great for load price models) and lots of useful fields such as temp, dew point, wind, pressure, cloud, visibility, precipitation. We would use it by mapping each node/zone to the nearest station(s) (or a weighted blend) and engineer features like Cooling/Heating Degree Hours, humidity/heat index, wind chill, and storm proxies. This dataset is very useful to our project because it provides all significant weather related information at a given location with vast historical information, allowing us to gauge the relationship between weather events and usage (and how it has changed over time).

The government provides this data free of charge and makes it very easy to access with a simple process in requesting an API key which is open to all. Using a python script, we have been able to pull over 100,000 rows of hourly weather data for the major ISOs in the United States.

2. PJM Interconnection - Dataminer (https://dataminer2.pjm.com/)

The PJM dataminer is a comprehensive ISO database containing usage, pricing, projection, and generation information about the grid in the PJM (Northeast) ISO region. We focus on PJM particularly because it is the region with the highest concentration of databases in the US and therefore has the highest likelihood of curtailment due to non-weather related signals. This resource supplements the usage data

provided by the EIA, however it provides a much greater level of detail. The EIA only provides 1 hour usage history and day ahead projected prices, while this database produces 5 minute real time pricing and load information. Additionally, the EIA only contains overall usage across the entire ISOs, while Dataminer gives us pricing information (known as LMPs) down to the node .

The main datasets that we intend on using are called Real-Time Five Minute LMPs, Instantaneous Load, and Generation by Fuel Type. The Real-Time Five Minute LMPs provide nodal pricing information with real time updates and substantial historical data. Pricing reflects usage, and can be used as a metric to assess how the correlation between weather related patterns and pricing has changed over the years, particularly at nodes associated with heavy data center usage. The Instantaneous Load is the most clear indicator of electrical usage at a given time, and also includes substantial historical data, which can be clearly used to measure changing trends between loads and their relationship to weather events. However, the Instantaneous Load is not recorded at the nodal level (just by subregion), so it is difficult to consider datacenter heavy nodes as they are diluted by the larger incorporated region, hence the need for nodal pricing. Generation by fuel type allows us to incorporate the impact of rising energy production sources that are directly tied to weather events, such as solar or wind, to account for when testing our hypothesis.

Accessing the database requires approval from PJM which was successfully requested, however, the site does not use a traditional API key. We have instead written a script which emulates the request made to display the pricing/load information shown on the web browser by making the same request via cURL. We have been able to successfully obtain over 2 million rows of data by making these requests via a simple python script, which times the requests to deliberately respect the usage limits.

Part 3: What Modules Are We Using?

Module 3: Visualization

We will use visualization to explore the relationships between weather patterns and electricity demand across the different regions. We expect to use this module in both the data exploration phase as well as the final report stage. We will create many time-series plots, scatter plots, as well as correlation heatmaps to compare demand patterns in data-center-heavy regions versus non-data-center-heavy regions. Due to one of our research questions focusing on the correlation between weather patterns and energy demand, visualization will be an integral part to detecting and drawing attention to shifts in demand. In our final report stage, visualization will be used to highlight our predictions and their fit to our data when utilizing different hyperparameters.

Module 6: Combining Data

We will apply data combination techniques to transform our many datasets into several datasets that contain the necessary data and format for visualization and analysis. During this step, we aim to increase data interpretability and to ensure consistency across sources, where the combinations of various data columns will be pre-prepared for analysis. While we are working with multiple high frequency datasets, we have already separated the datasets by location and thus are focusing on ensuring the common column of “time” is where the datasets are to be merged in a many-to-many relationship. This step is crucial in the beginning stages of our projects as we need our input data to be suitable to be run through our code whether for initial or later visualization as well as for analysis. We expect to use this module primarily during the data cleaning stages of the project before moving into visualization and model development.

Module 8: Prediction and Supervised Machine Learning

We will construct supervised learning models to predict electricity demand in a region using weather variables. We will then compare the models’ performance across regions with high and low data-center concentration and evaluate the difference in prediction accuracy. Since one of our research questions revolves around model accuracy, creating and training our own prediction model is essential for

evaluating whether standard energy usage prediction models perform worse depending on the concentration of data-centers in a specific region. Some concepts that will be used are train-test split, calculating the Mean Squared Error (MSE), linear regression etc. We expect to use this module during the data-analysis stage of the project to understand any underlying relationships.

Part 4: Preliminary Results and Methods:

Data and Scope

This prototype analyzes weather–price relationships in PJM. The export has 1,829 records across 49 nodes and 7 regions, covering 2021–2025 (earlyAI/recentAI), with target lmp and weather variables CDH, HDH, TEMP C, RH PCT, WIND MS, DEW C, SLP HPA, and PRECIP MM.

Methods

Results were generated with the following data-acquisition and processing pipeline:

1. Energy data source (PJM): We used the PJM bundle containing day-ahead LMP parquet files (pjm da full system parquet), load forecast files, and node metadata (feature data). The bundle was downloaded from the shared Box link and placed at data/pjm data (or accessed via JM DATA ROOT).
2. Weather data source (NOAA ISD): For each selected PJM node, we mapped the node to its nearest NOAA ISD station, then downloaded hourly "global-hourly" weather CSV files for years 2021–2025. Raw weather files were saved under data/noaa isd global hourly/(stationId)/.
3. Parsing and cleaning: NOAA coded fields were converted to numeric units: temperature/dew point (tenths of ◦C to ◦C), sea-level pressure (tenths of hPa to hPa), wind speed (tenths of m/s to m/s), and precipitation (tenths of mm to mm). Sentinel missing values (e.g., 9999/99999 patterns) were set to null. PJM timestamps were parsed as UTC; NOAA timestamps were parsed and localized to UTC.
4. Feature engineering: We computed CDH and HDH using a base temperature of 18◦C and computed relative humidity (RH PCT) from temperature and dew point using a Magnus-form approximation.
5. Merge logic: For each node-year, PJM LMP rows were merged with NOAA hourly rows using nearest-time matching in UTC (merge asof) with a 30-minute tolerance. Rows without matched weather values were excluded from correlation calculations.
6. Analysis construction: For each node-year-variable pair, we computed Pearson correlation with day-ahead LMP, then stored results with identifiers (nodeId, regionId, year, period, variable, target). Period labels were assigned by year rule in code (earlyAI for 2021 and recentAI for 2022+ in this dataset slice).
7. Visualization aggregation: The two report figures were generated by grouping correlations by data-center bucket and variable/year, plotting group means, and adding standard-error bands/bars for uncertainty context.

Preliminary Results

Fig. 1 is the primary comparison of correlation shifts (recentAI - earlyAI) for data-center-heavy vs non-heavy nodes.

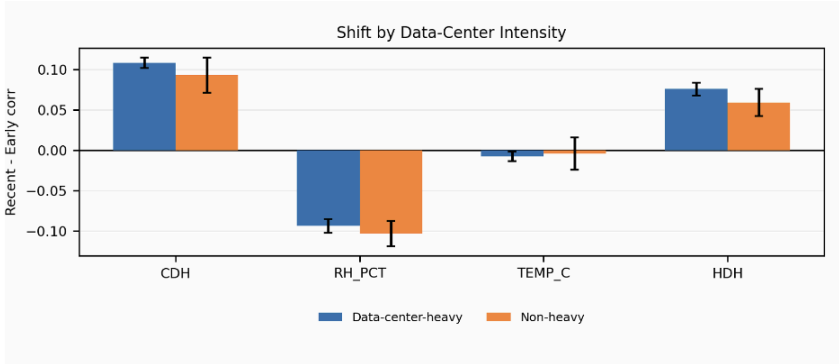

Figure 1: Mean shift in weather-to-LMP correlation (`recentAI` - `earlyAI`) by data-center intensity bucket, with standard-error bars.

Figure 1 represents how the weather–LMP relationship changed between periods for each bucket. Positive values indicate stronger recent correlation; negative values indicate weaker recent correlation. In our preliminary sample, CDH and HDH increase for both buckets (slightly more in the heavy bucket), while RH PCT becomes more negative for both buckets (slightly more in the non-heavy bucket). This indicates measurable temporal shifts, but not a uniform "heavy regions always weaken more" effect.

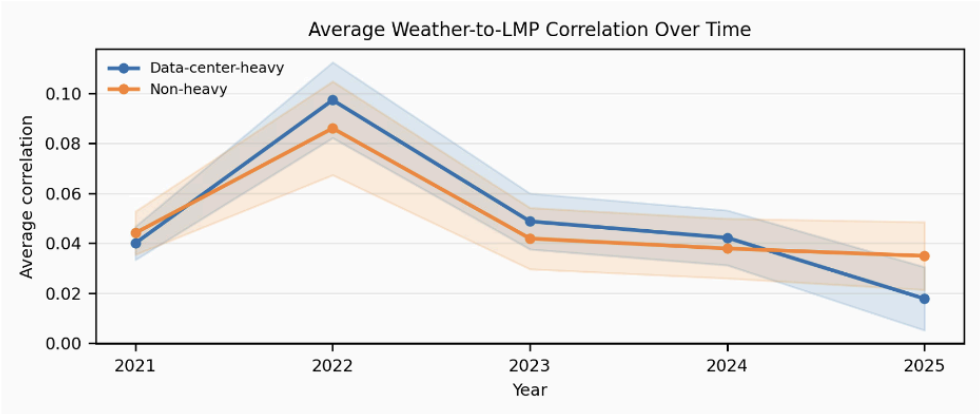

Figure 2: Supplemental trend view: yearly mean weather-to-LMP correlation by bucket (shaded bands are standard error).

As a supplement, Fig. 2 shows that both groups follow a similar time pattern (rise to 2022, then decline through 2025), with only small year-to-year gaps between heavy and non-heavy nodes.

Significance to Research Question

Our research question asks whether data-center-heavy areas exhibit stronger or systematically different weather–price relationships than non-heavy areas. The primary result (Fig. 1) provides partial evidence of difference by variable (e.g., slightly larger CDH/HDH increases in heavy nodes), but not a consistent directional gap across all variables. The supplemental time-series result (Fig. 2) reinforces this interpretation: both groups move similarly over time, suggesting common system-wide effects, with only modest group separation. Together, these preliminary findings support the project's central hypothesis that weather sensitivity is changing over time, while also showing that the "data-center-heavy vs non-heavy" distinction is nuanced and requires deeper modeling in later

sections.

Part 5: Reflection and Next Steps:

Successes/Mostly Complete
For our final report, we are closing in on answering the first research question, although the results at this stage don't draw any clear conclusions with conflicting results. We will have to perform more investigations and model the data differently to understand if weather is becoming less related to energy usage in data center heavy regions. Data gathering, cleaning, aggregation, correlation processing, frontend development, and placeholder pages for the remaining stages of the project are all fully developed.

Challenges/Incomplete
Tasks that have so far been incomplete are mostly related to our second research question, which requires training a predictive model primarily leveraging weather features to analyze its efficacy in predicting energy usage/pricing in data center heavy vs normal regions. With our comprehensive list of cleaned data, we should be able to easily featurize the data and train several models for comparison.

Collaboration plan reflection
We are following the initial plan quite closely with the research question not yet disproven by the data we have been able to gather. We separated the code repository with data storage as github could not host all of the data given its massive size.

Next Steps
Our next steps will be to utilize machine learning and feature selection to find a selection of data columns that gives the best prediction, then to analyze this prediction using comparative scripts and display our results in the React webapp using useful graphics and tables.

Part 6: AI Disclosure:

Artificial intelligence Tools: Cursor
Code Generation: All code for this project was generated using Cursor (Auto mode). A comprehensive list of prompts is available here:
(https://github.com/Jaiparmar940/energy-weather-app/blob/main/docs/USER_PROMPTS_LOG.md).
Document formatting: Latex code for section 4 was written by cursor to properly format tables, variable names, and ensure the accuracy of the statements made (and correcting where appropriate). The corresponding Latex output was pasted into docs while figures were inserted as screenshots

Supplemental Information:

Github code: https://github.com/Jaiparmar940/energy-weather-app
Duke Box data: https://duke.box.com/s/i75cw52pljlceupqarzp2vasaa31sgf7